

Causal inference for marketing

a PyMC-ecosystem cookbook

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This is the uplift/targeting technique taken to real analytical depth. It is the same business problem as the introductory worked example — a €10 discount email (cost $\approx$ €8 per contact), and the question *for whom does it pay?* — but here we **interrogate** the analysis along the three axes that separate a result from an analysis: (1) do the estimators actually work, and how do they fail; (2) is the causal claim identified, and how fragile is it; (3) what is the decision worth, in euros, and how sensitive is it to our assumptions. Every number and figure below was executed in PyMC on 1,400 simulated customers with a known ground-truth effect.

Depth 1 — Estimator bake-off and failure modes

The estimand is $\tau(x) = \mu_1(x) - \mu_0(x)$, the difference of two response surfaces. There are three standard ways to model it, and they are *not* interchangeable:

- **S-learner** — one model with treatment as an extra feature, $f(x, T)$; effect = $f(x, 1) - f(x, 0)$.
- **T-learner** — two separate models, one per arm; effect = $\hat{\mu}_1(x) - \hat{\mu}_0(x)$.
- **Bayesian Causal Forest (BCF)** — one model, structurally split into a *prognostic* part (with the propensity fed in) and a separately-regularized *treatment* part, $y = \mu(x, \hat{e}(x)) + \tau(x) T$.

We score each on **PEHE** (root-mean-square error of the individual effect — the HTE accuracy metric that matters), rank **correlation** with the truth, **ATE bias**, and **90% interval coverage** (is the uncertainty honest). Run on both a randomized past campaign and an observational one:

regime	estimator	PEHE ↓	corr ↑	ATE bias	90% coverage
randomized	S-learner	9.21	0.73	+0.62	0.73
randomized	T-learner	4.67	0.91	+1.30	0.91
randomized	BCF	5.49	0.87	+1.35	0.86
observational	S-learner	9.48	0.72	−0.65	0.73
observational	T-learner	4.70	0.90	+0.17	0.90
observational	BCF	5.23	0.87	−0.16	0.84

The S-learner has a genuine failure mode, and it is consistent across regimes: its PEHE is roughly **double** the others, and — worse for decision-making — its intervals cover the truth only **73%** of the time when they claim 90%. The mechanism is structural: with the treatment as one feature among six, BART's regularization prunes the single treatment split into near-irrelevance,

so the model *flattens* the heterogeneity toward the average. It doesn't just add noise; it systematically under-separates who responds — and then reports over-confident intervals about it.

T-learner and BCF are both good, and comparable here. That is itself an important, non-obvious result: BCF does not dominate. The reason is diagnostic — in this setup the confounders are all *observed*, so simply conditioning on x (which the T-learner does by fitting each arm on x) already removes the bias. BCF's extra machinery (propensity in the prognostic term, separate shrinkage on τ) is *insurance* that pays off in the harder regimes — poor overlap, strong targeted selection, unobserved confounding — not a free lunch when the problem is easy. We stress-test exactly that in Depth 2.

Calibration is not a single number. A headline “90% coverage” can hide intervals that are honest in the bulk and dishonest in the tails — which is exactly where targeting decisions live. Checking coverage *within deciles of the true effect*:

Coverage holds near 0.9 across the middle but **sags at both extremes** — the strongest persuadables and the worst sleeping-dogs have their effects pulled toward the mean, so the true value slips outside the interval more often. The practical consequence is honest and useful: near the decision boundary (effects around the €8 cost) the intervals are trustworthy, so the targeting rule is safe; but you should expect the model to *under-state* the very best customers, not over-sell them.

Depth 2 — Identification: confounding and how fragile the claim is

The randomized regime was a warm-up. The realistic case is **observational**: historically, marketers emailed the customers they liked — the engaged, the recently-active. Treatment is now *targeted selection*, and the naive analysis breaks.

The bias is large and has a direction. A plain difference in means gives an estimated ATE of **€10.83**, against a true **€5.80** — the email looks nearly **twice** as effective as it is (left bar of the figure above). The reason is pure confounding: the emailed group was already going to spend more, and the naive comparison credits that pre-existing gap to the email.

Adjustment works — but only because overlap holds. Conditioning on x (T-learner/BCF) recovers the truth here, but that recovery has a precondition — **positivity**: at every kind of customer there must be both emailed and non-emailed people, or there is no like-for-like comparison to make and the model is silently extrapolating.

Propensities run ≈ 0.08 – 0.93 with both groups present throughout, so adjustment is valid. Had the marketing team emailed *only* the highly-engaged (propensities pinned near 1 for that group), no amount of modelling could recover their untreated behaviour — you would be inventing a counterfactual with no data behind it. This diagnostic is the first thing to run on any observational uplift job.

The part you cannot test — and how to price it. Unconfoundedness (“we measured every confounder”) is an assumption, never provable from the data. The honest move is to ask *how strong an unmeasured confounder would have to be to overturn the decision*. We inject a hidden confounder U that drives both emailing and spend, sweep its strength, and adjust for the observed x only:

The tipping point is uncomfortably close: a hidden confounder just $\approx 0.25\sigma$ strong (a quarter of the outcome noise) is enough to inflate the estimated ATE past the €8 cost, at which point the analysis would wrongly endorse *emailing everyone*. That is the quantitative version of “these results assume no important hidden confounder” — and it says the honest recommendation here is not “trust the model” but “the decision is only moderately robust; validate the high-value segment with a small randomized holdout before scaling.” **This is the sentence that separates a credible causal consultant from a dashboard.**

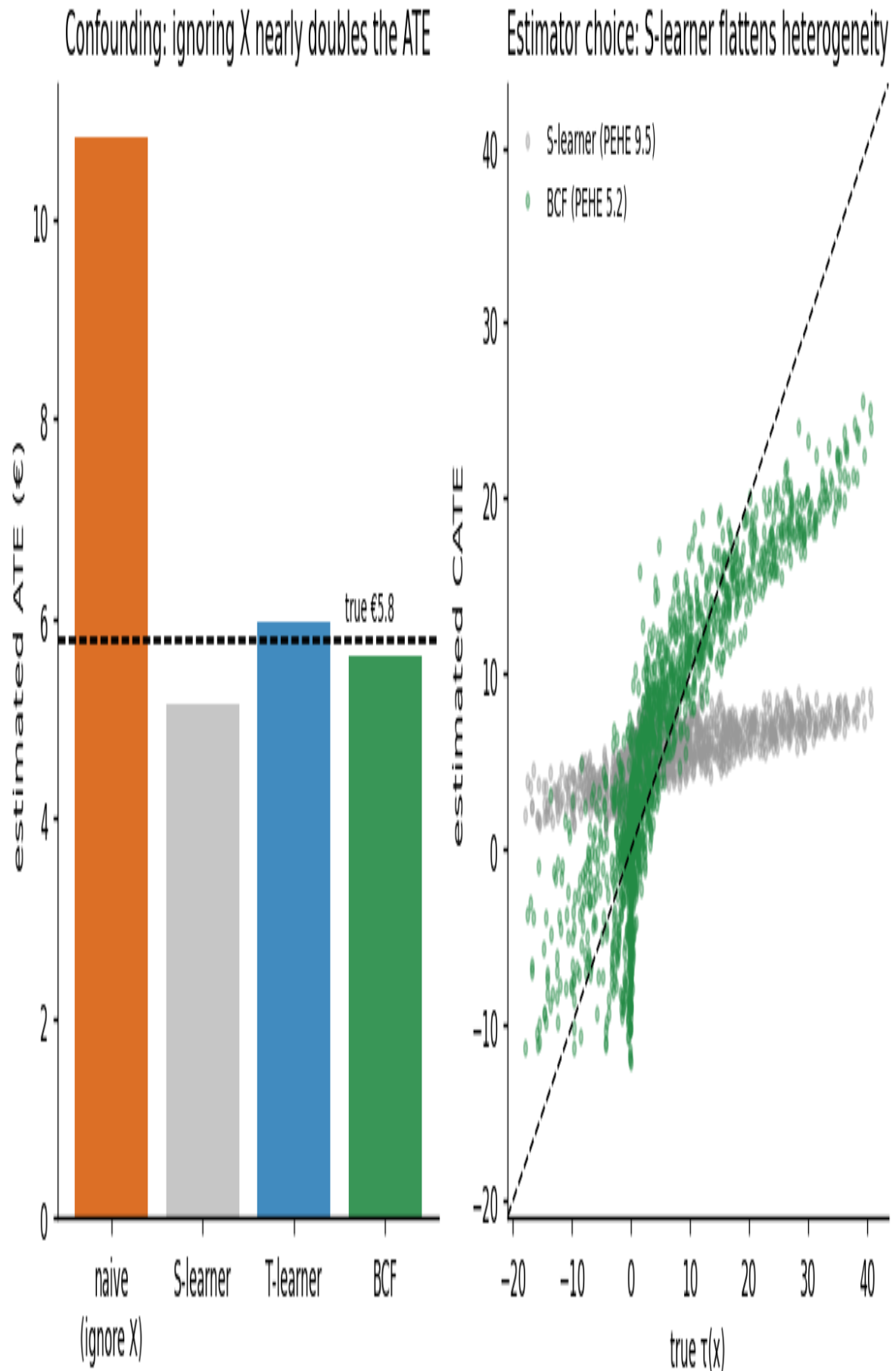


Figure 1: Left: the average effect estimated four ways — ignoring the covariates nearly doubles it (confounding, Depth 2); the flexible adjusted methods land near the truth. Right: individual-effect recovery — the S-learner compresses the true spread toward the middle, while BCF tracks the diagonal.

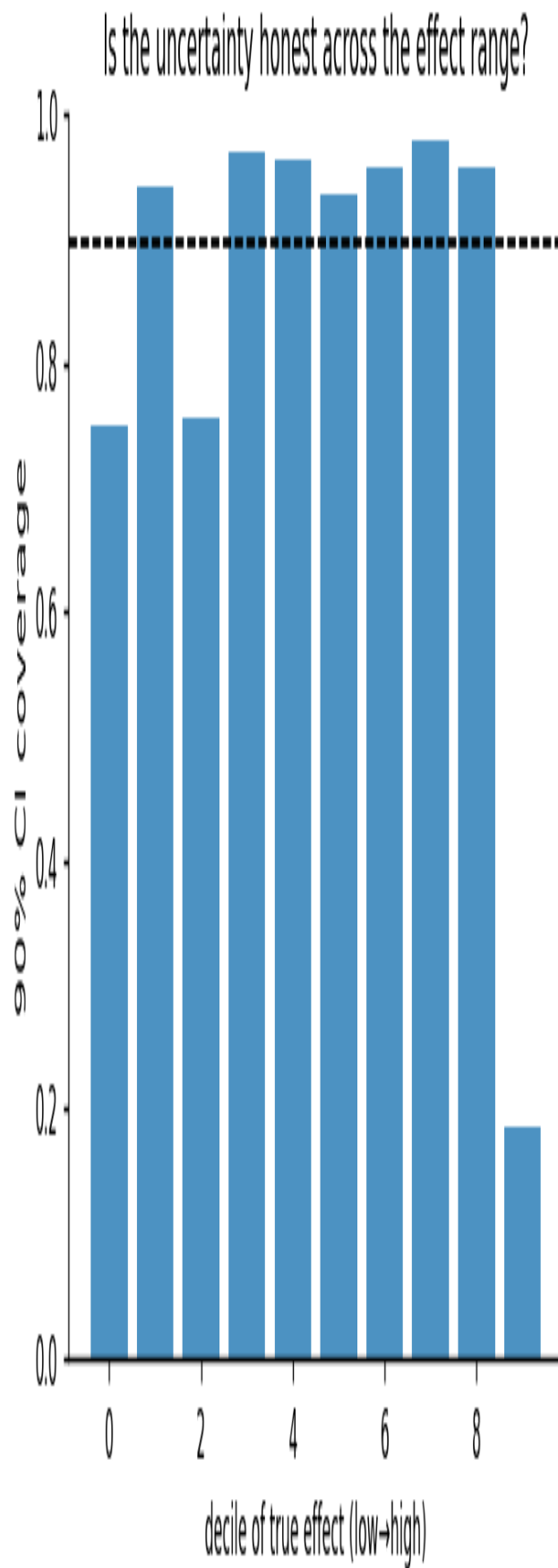


Figure 2: 90% interval coverage within each decile of the true effect. Honest in the middle; it sags at the extremes, where BART shrinks the largest positive and negative effects toward the mean.

Overlap check: do both groups exist everywhere?

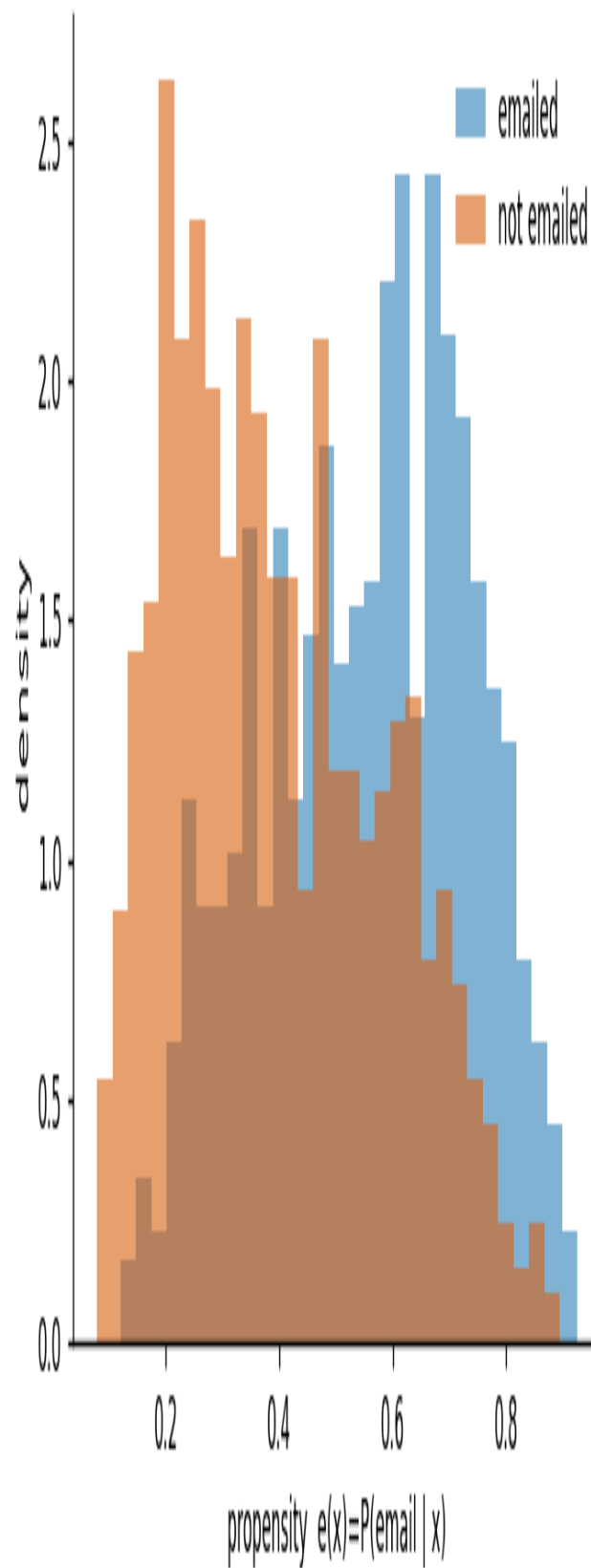


Figure 3: Estimated propensity by group. The two distributions span the whole range (≈ 0.08 – 0.93) with no customer pinned at 0 or 1 — overlap holds, so covariate adjustment is legitimate rather than extrapolation.

Sensitivity: how much hidden confounding overturns the call?

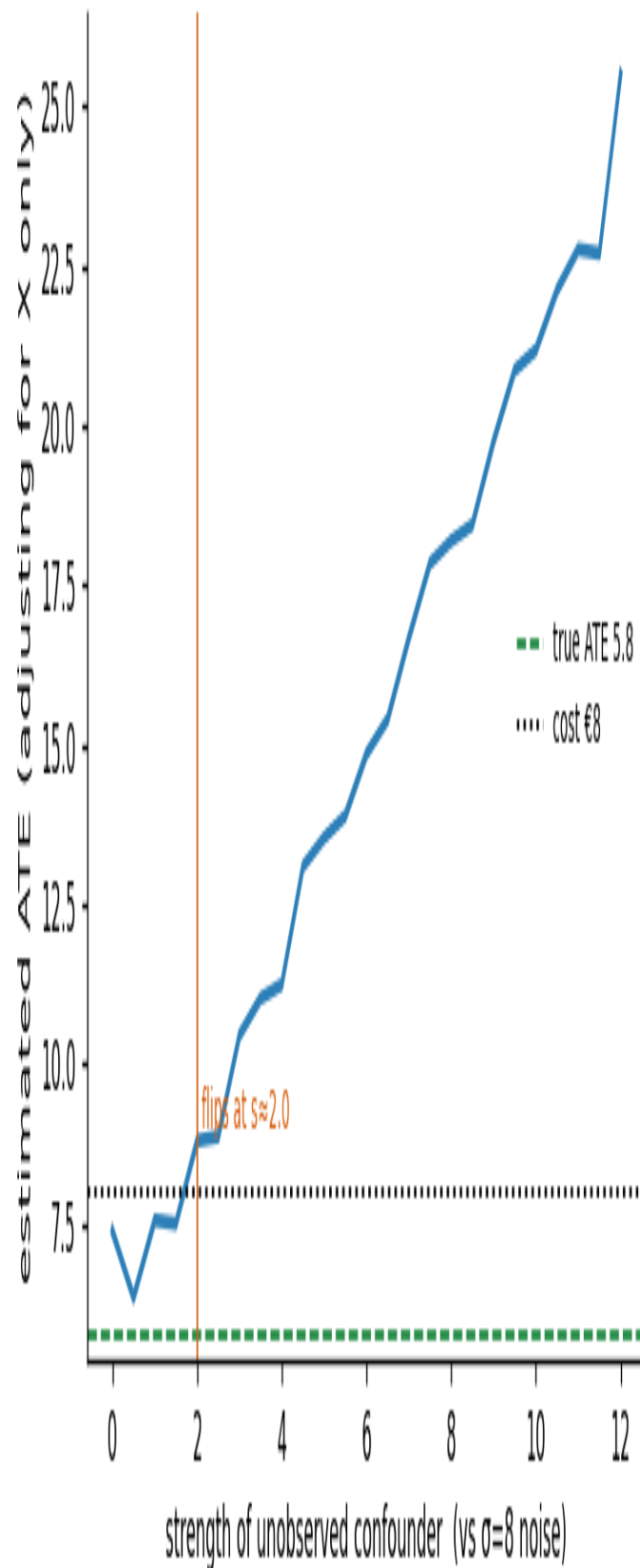


Figure 4: Estimated ATE (adjusting for observed features only) as a hidden confounder strengthens. A confounder only $\approx 0.25\times$ the outcome noise pushes the estimate past the €8 cost line — at which point the wrong conclusion (“emailing everyone pays”) becomes attractive.

Depth 3 — The decision, in euros, and its sensitivities

Turn the (BCF, observational, adjusted) per-customer effects into money. Rank customers by estimated uplift, contact from the top, pay €8 each.

The headline decision. Emailing everyone **loses €3,087** — the sure-things' giveaways and the sleeping-dogs' *negative* uplift outweigh the persuadables. Targeting the model's top segment ($\approx 29\%$ of the base) makes **+€4,502**, which is **96%** of the €4,679 an oracle with perfect knowledge of every true effect could achieve. The swing from $-\text{€}3.1\text{k}$ to $+\text{€}4.5\text{k}$, by contacting under a third of the base, is the entire business case — and it exists only because the effect is heterogeneous.

Sensitivity to the cost assumption (middle panel). The whole policy pivots on c , the true margin cost of a redemption. As c rises the optimal profit falls *and* the target set shrinks — at a high enough cost only the strongest persuadables clear the bar. The lesson for the room: the single most important number to get right is not in the model, it's the **finance input** c ; a wrong cost silently mis-sizes the whole campaign.

Sensitivity to the confidence bar (right panel). Contacting where $P(\tau > c)$ exceeds a threshold has an interior optimum — too strict and you leave profitable persuadables unmailed; too loose and you mail customers whose effect you can't distinguish from the cost (or from *negative*). The Bayesian posterior is what makes this knob available at all; a point estimate has no "how sure" dimension to tune.

Pricing the uncertainty — value of information. The remaining gap to the oracle is the *cost of not knowing*. We can price it: the option value of resolving each customer's effect is $\mathbb{E}[\max(\tau - c, 0)] - \max(\mathbb{E}[\tau] - c, 0)$, summed over the base. It comes to **€461**, and **€448 of it sits on the 686 "straddler" customers** whose credible intervals cross the €8 line — the ones we genuinely can't classify. That number is directly actionable: it is the **upper bound on what a follow-up experiment is worth**, and it tells you *whom* to put in it (the straddlers, not the whole base). A rational next step is a small randomized test scoped to those 686 — and now you can justify its budget to the euro.

What transfers to the lecture

Three lessons, each defensible because we watched them happen rather than asserted them:

1. **Adjust for confounders or your effect roughly doubles** (€10.8 vs €5.8) — and *check overlap* before you trust the adjustment, because adjustment without positivity is extrapolation in disguise.
2. **The estimator is a decision, not a detail** — the S-learner quietly flattens heterogeneity and lies about its own uncertainty (73% coverage for a "90%" interval); a targeting policy built on it mis-ranks who to contact.
3. **Price the uncertainty instead of hiding it** — the credible intervals aren't decoration: they set the confidence bar, they flag the fragile-to-confounding decision, and they put a **€461 price tag** on the next experiment. That is the concrete, monetised answer to "why Bayesian," delivered without ever saying the word.

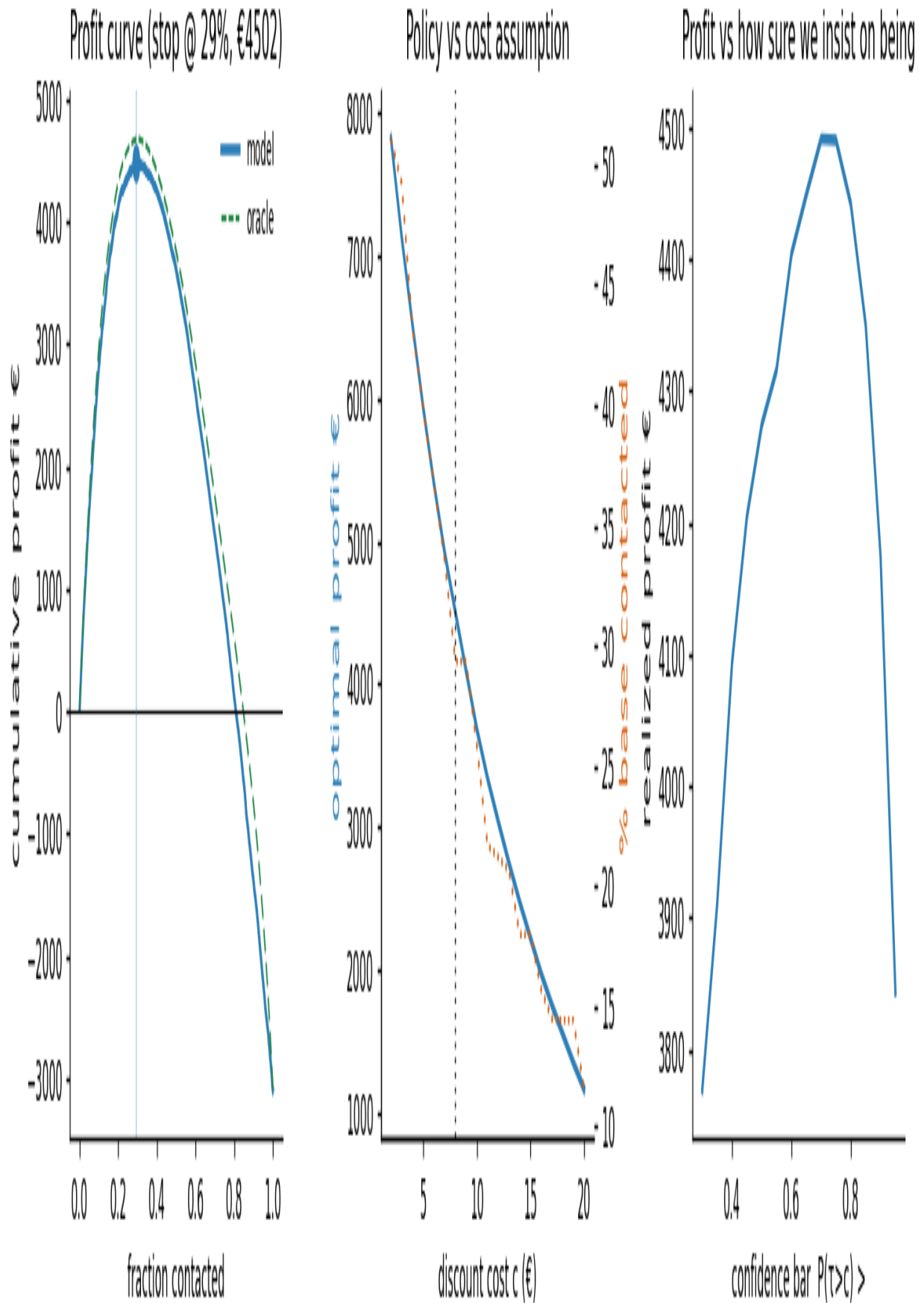


Figure 5: Left: cumulative profit — email-everyone loses money, the model policy captures ~96% of the oracle. Middle: the optimal policy as a function of the assumed cost c (profit falls, and you contact fewer people, as the discount gets more expensive). Right: realized profit as a function of how confident we insist on being before contacting.